

Pattern Formation and Positional Memory in Hydra Regeneration Models

Szymon Cygan · Anna Marciniak-Czochra · Finn Münnich · Moritz Mercker
Heidelberg University, Heidelberg, Germany



1 Motivation

1.1 Hydra as a patterning problem



Hydra regenerates a head after amputation, forms a single head, and remembers the original polarity of its tissue. Each head is built around a head organiser marked by Wnt/ β -catenin signalling: during regeneration, Wnt3 becomes localised at the tip of the future head. Behind this behaviour lie three different phenomena:

- regeneration,
- pattern selection,
- positional memory.

Activator–inhibitor models already explain regeneration and pattern selection, but positional memory requires an additional variable: source density.

2 What the activator–inhibitor model already does

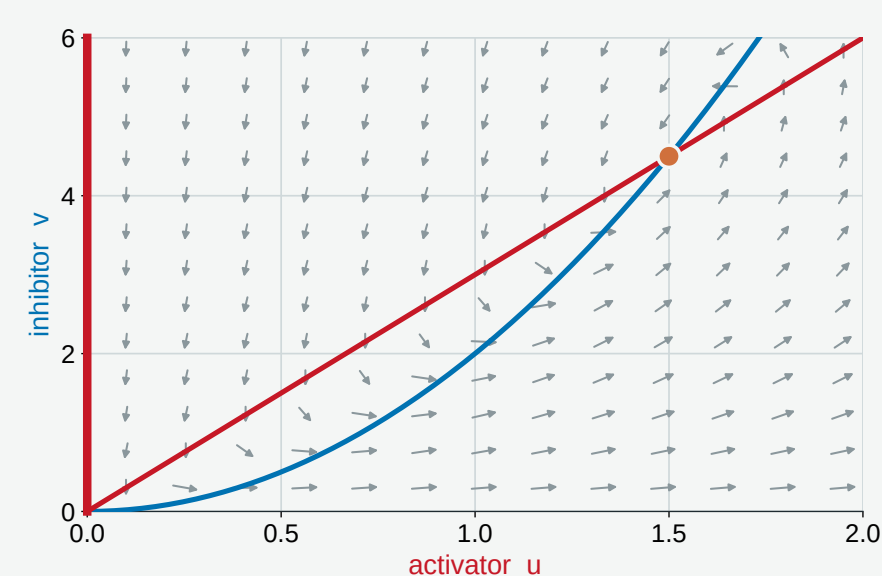
2.1 Production terms change the phase portrait

$$\begin{aligned}\partial_t u &= D_u \Delta u + \frac{au^2 + q_u}{v} - \mu_u u + p_u, \\ \partial_t v &= D_v \Delta v + bu^2 - \mu_v v + p_v,\end{aligned}\quad x \in [0, L], \quad t \geq 0$$

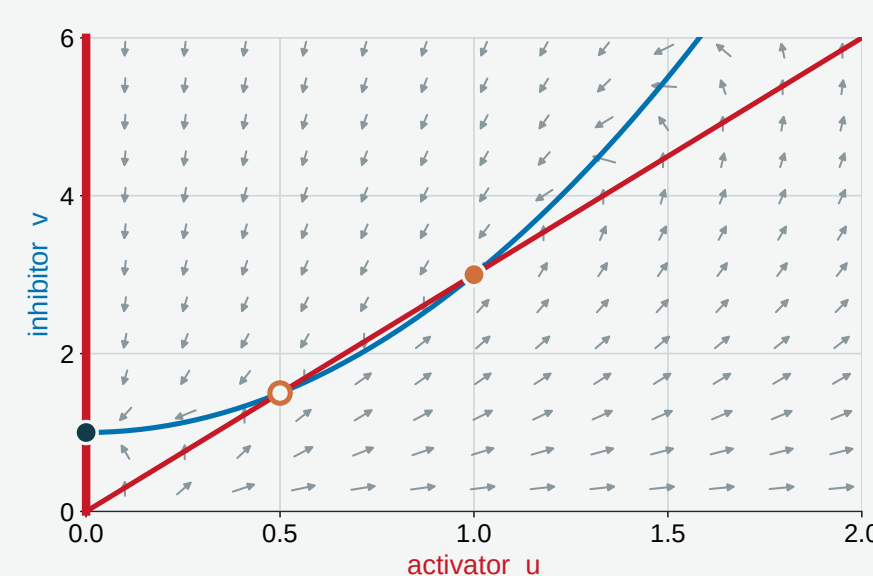
- u : activator, short-range and self-enhancing,
- v : inhibitor, long-range, produced by u .
- p_v : basal inhibitor production,
- p_u : external activator input,
- q_u : activator input under inhibitor regulation.

Nullclines

— u -nullcline — v -nullcline ● stable rest ○ unstable threshold ● active state (Turing candidate)



1 steady state ($p_v = p_u = q_u = 0$)
no resting state: an arbitrary perturbation leads to a pattern.



3 steady states ($p_v > 0$)
small perturbations vanish back to the rest state $u = 0$.

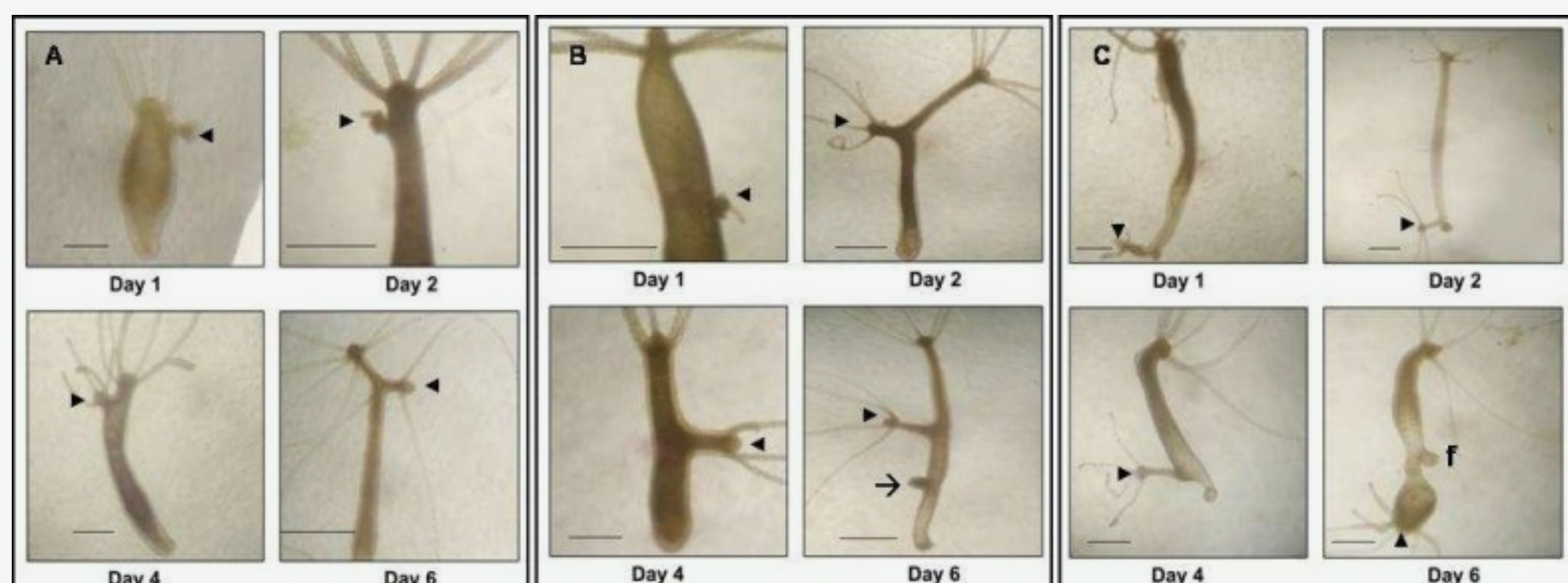
2.2 Inhibitor production is essential

$p_v = 0$: Only one steady state, which can be Turing unstable. Any perturbation forms a pattern.

$p_v > 0$: A stable rest state, an activation threshold and a pattern-forming state.

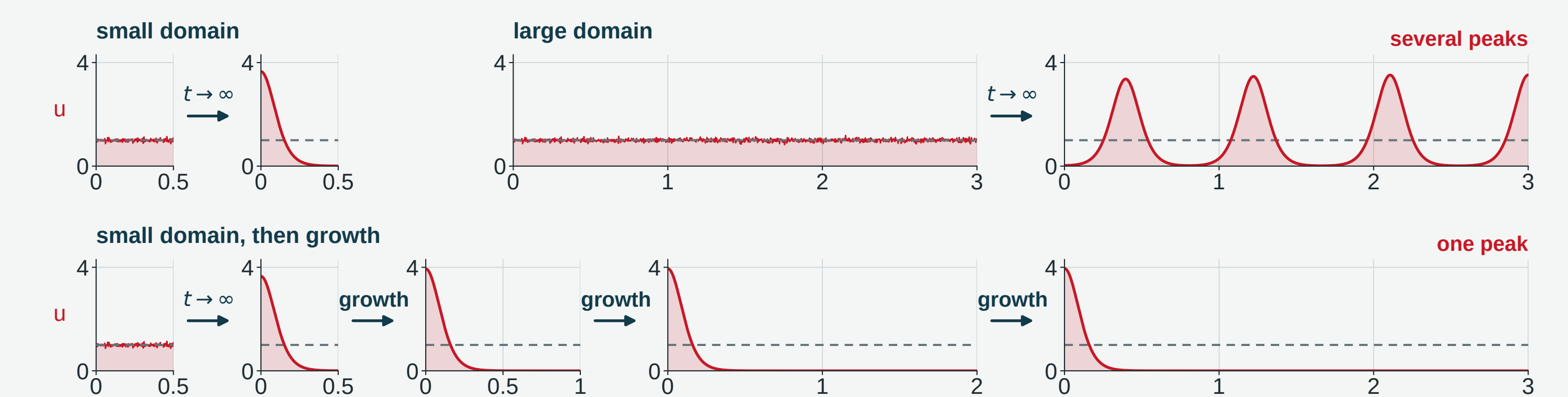
For everything that follows it is essential to have more than one steady state.

2.5 Grafts - the model chooses the position



Kadu, V., Ghaskadbi, S. S. & Ghaskadbi, S. (2012). *Int. J. Mol. Cell. Med.* 1(1), 11–20. CC BY 3.0.

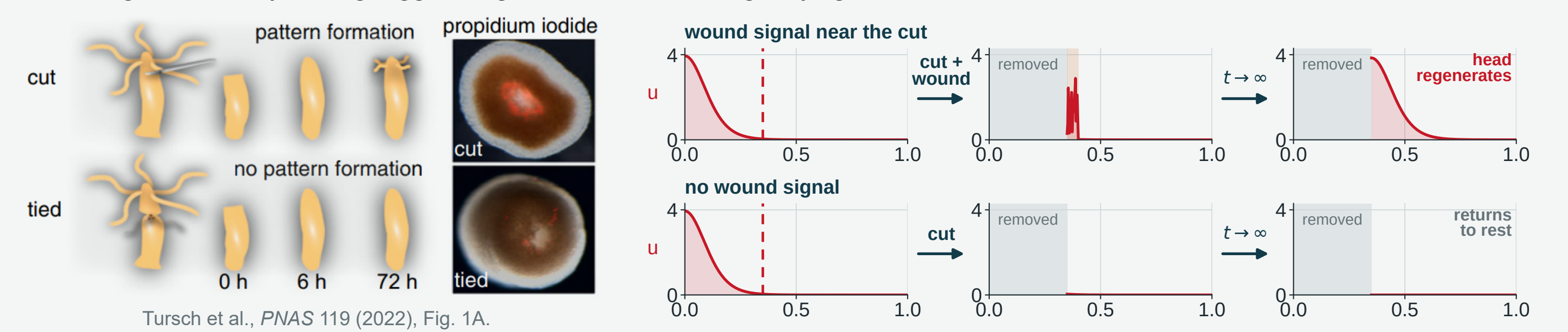
2.3 Pattern selection depends on history



Starting near the diffusion-unstable state E_+ , a small domain forms one peak and a large domain several. Growing the small domain to the same large size keeps a single peak.

2.4 Regeneration needs a wound

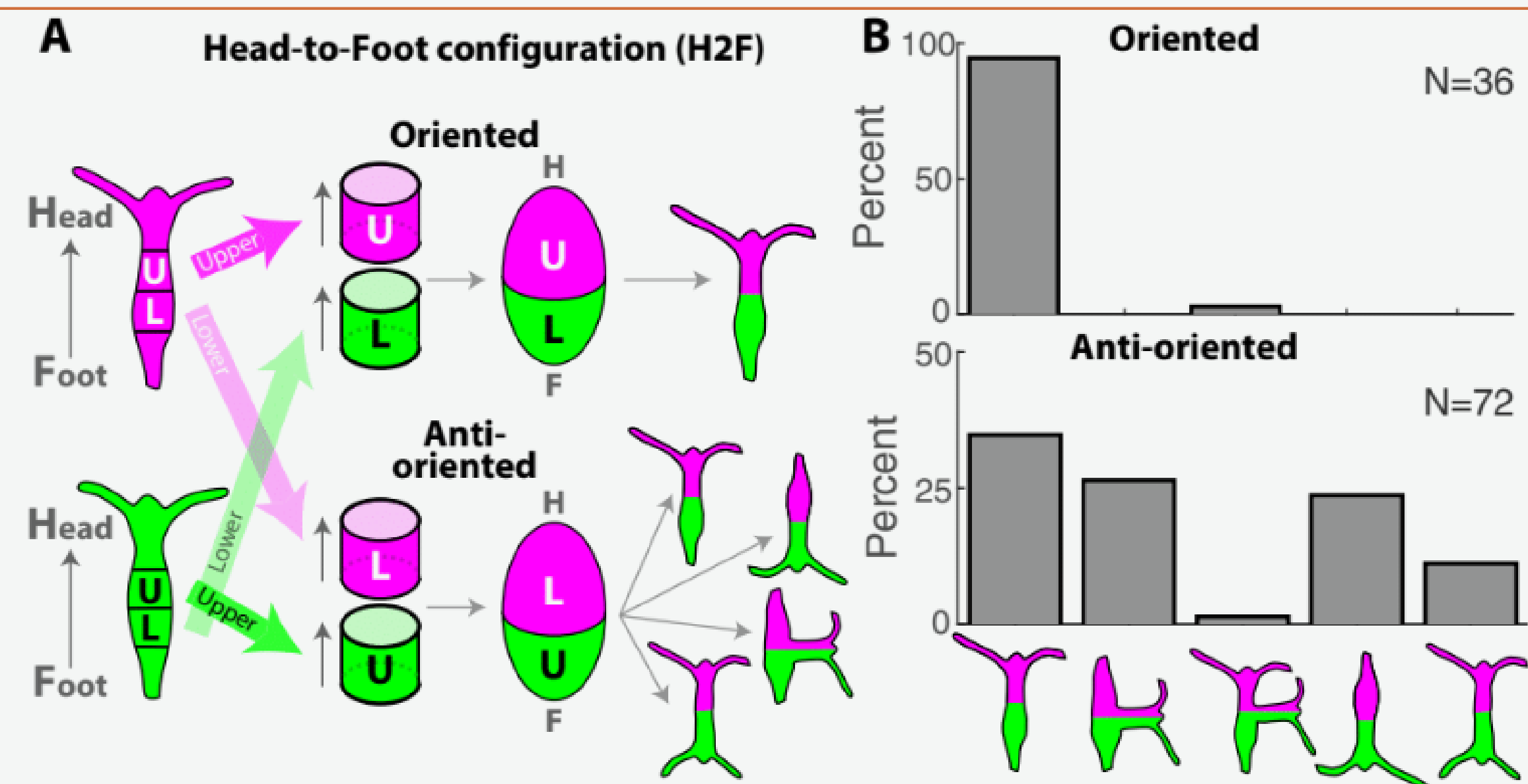
Removing the head by cutting triggers regeneration. Removing it by ligation, without a wound, does not.



Bistability is essential. With a single steady state regeneration always occurs. With three steady states the system is bistable, the cut tissue returns to rest and only a large perturbation from the wound brings back a pattern. Biologically, this perturbation is the wound response. Injury triggers Ca^{2+} and ROS signalling, which activates MAPKs and up-regulates Wnt3 at every cut, independent of its position (Tursch et al., 2022).

3 Positional memory and source density

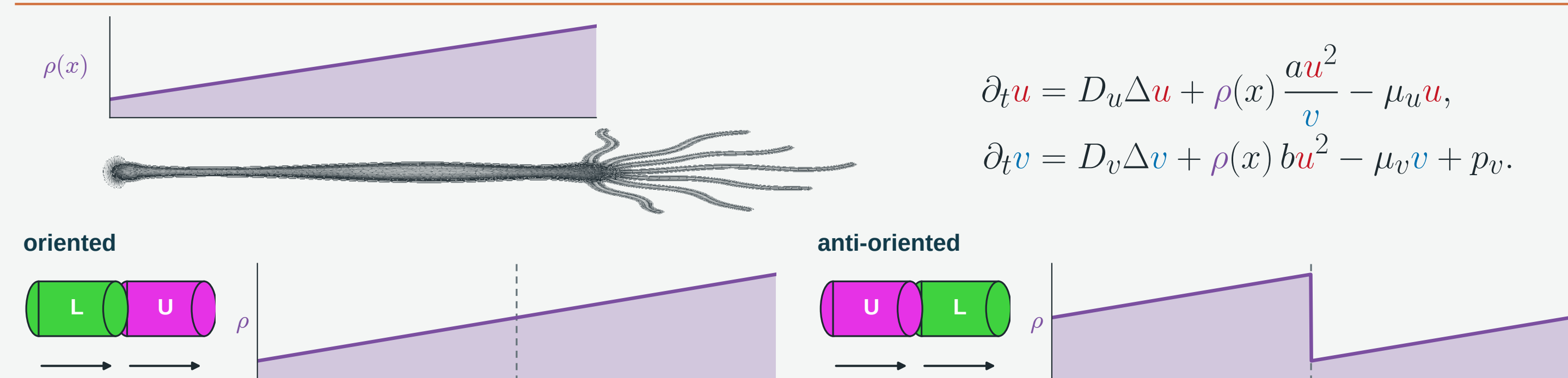
3.1 Grafting exposes missing information



Head-to-foot grafts, oriented vs. anti-oriented. Livshits et al. (2022). K. Keren group.

Reversing the orientation of comparable tissue pieces changes the distribution of outcomes. Near the resting state, u and v alone do not distinguish an upper-body fragment from a lower-body one.

3.2 Prescribed $\rho(x)$ as tissue competence

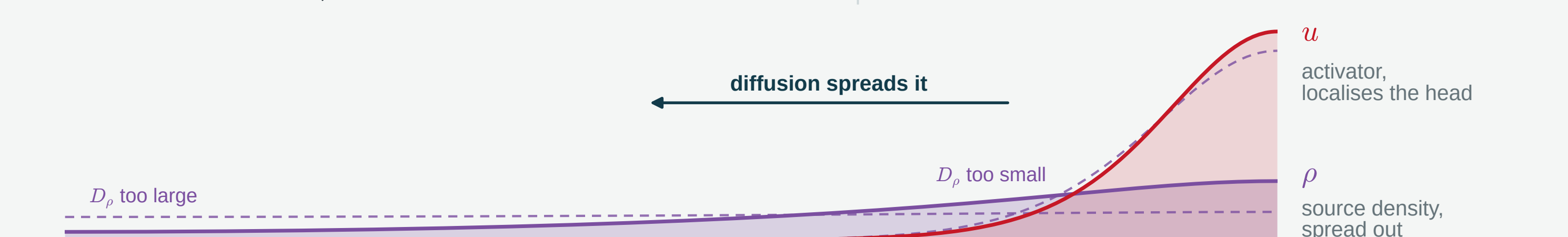


A fixed source-density profile encodes tissue origin before the activator is present and biases where a wound-triggered peak forms (source-gradient idea, Suzuki & Takagi 2011).

3.3 Dynamic source density model

$$\begin{aligned}\partial_t u &= D_u \Delta u + \rho \frac{au^2}{v} - \mu_u u, \\ \partial_t v &= D_v \Delta v + b\rho u^2 - \mu_v v + p_v, \\ \tau \partial_t \rho &= D_\rho \Delta \rho + u - \rho.\end{aligned}$$

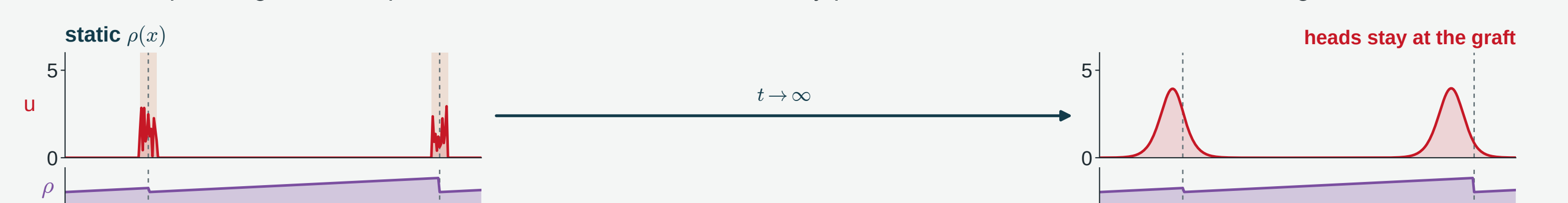
- $\rho(0, x)$ stores the grafted configuration,
- τ sets the lifetime of memory,
- D_ρ communicates positional information in space.



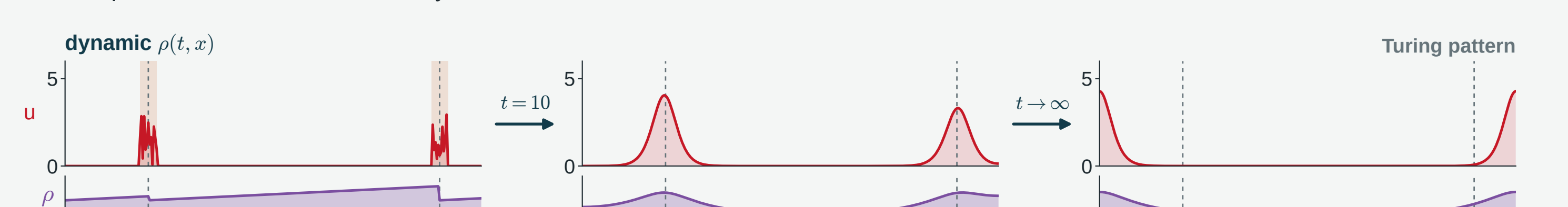
D_ρ too small: memory stays sharply localised. Intermediate: a usable gradient forms. Too large: spatial information is erased.

3.4 Static vs dynamic source density - metastability of patterns

Three tissue pieces generate a piecewise-linear initial source-density profile. Wounds are induced at the two graft interfaces.



Static source density $\rho(x)$: activator peaks form near the wound sites and remain there. These positions are not Turing-selected, but the two-peak state is stable and may mark sites of new-axis formation.



The source density is initialised as $\rho(x, 0) = \rho_0(x)$, with large τ chosen so that its dynamics is slow relative to u and v . On the fast activator-inhibitor timescale, peaks form near the graft interfaces as in the static case. This graft-pinned configuration is metastable and, as ρ relaxes, eventually destabilises towards the Turing-selected two-head pattern.

Source density is not needed to make a head. It may be needed to remember where a head should form.